

Retrieval Augmented Generation (RAG) & Embeddings

Making LLMs Smart with Your Private Data

Agenda

1. Why LLMs alone are not enough
2. What is RAG?
3. How RAG works
4. Use cases
5. What are Embeddings?
6. How similarity works
7. Popular embedding models
8. Trade-offs

The Core Limitation of LLMs

LLMs can't see your private data

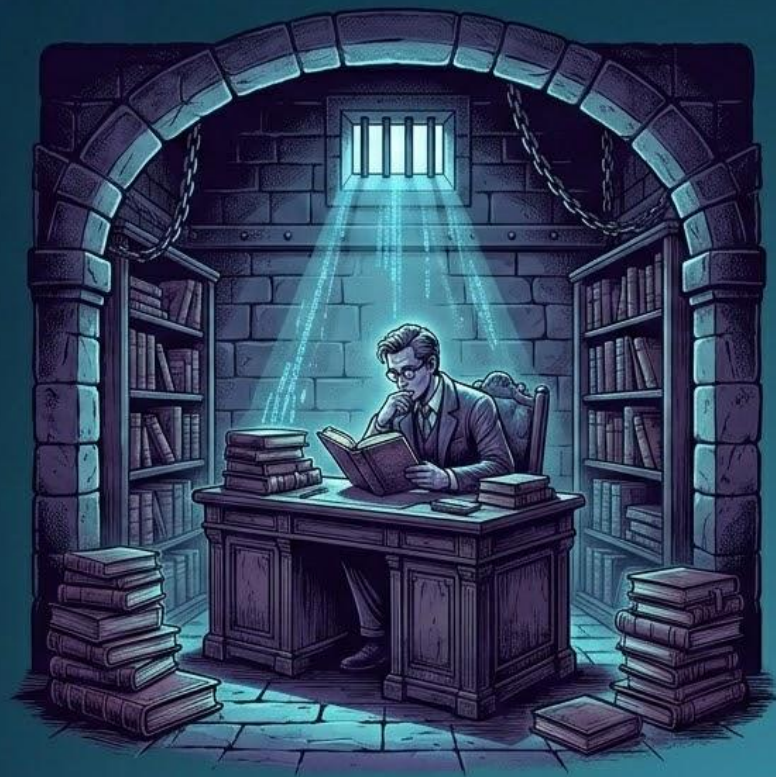
- Trained on public data
- No access to company documents
- Hallucinations on internal knowledge



The Brain in a Windowless Room Problem

The Windowless Room Problem

- Knowledge is confined to a fixed database
- No access to new external or private data
- Cannot break out of the pre-trained boundaries
- Analogy for limited model context



What is RAG?

Retrieval Augmented Generation



RETRIEVAL

Find relevant information from external data sources



AUGMENTED

Enhance the LLM's prompt with retrieved knowledge



GENERATION

Produce accurate answers based on the complete context

RAG provides LLMs with access to internal and real-time private documents to reduce hallucinations and provide accurate information.

How RAG Works (4 Simple Steps)



RAG Architecture



Real-World Use Cases

Leveraging RAG for enhanced productivity



Internal Docs



HR Policies



Product Manuals



Financial Reports



Customer Support



Legal Contracts

WHAT IS AN EMBEDDING?



An embedding is a technique for representing categorical variables (like words, objects, or users) as dense vectors in a continuous, high-dimensional space. Features are learned through model training.

Why Vectors?



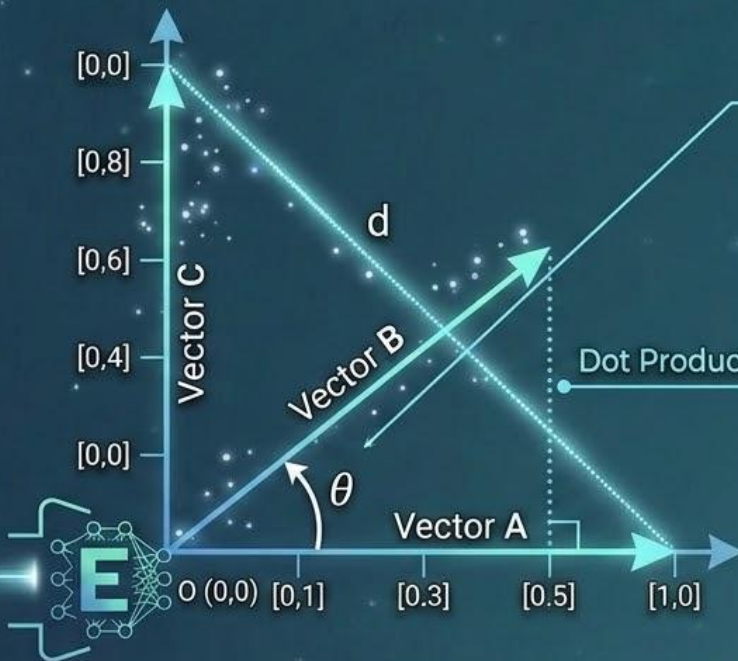
Giving computers a Semantic GPS

Embeddings map the meaning of words, sentences, or concepts into a multi-dimensional mathematical space.

Vectors are the precise coordinates that allow computers to navigate and understand semantic relationships, enabling powerful search and reasoning.

How Do We Measure Similarity?

Vector metrics quantify the geometric closeness or directional alignment of data embeddings.



Cosine Similarity ($\cos \theta$)

$$\text{Sim}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

-1 to 1 +1 to 1
Focuses on directional orientation, ignoring magnitude.

Dot Product ($A \cdot B$)

$$A \cdot B = \sum (a_i * b_i)$$

Algebraic product of magnitude and alignment.

Euclidean Distance $d(A, C)$

$$d = \sqrt{\sum (a_i - c_i)^2}$$

Straight-line physical distance.

Popular Embedding Models

 Model	 Dimensions	 Strengths	 Best For
OpenAI text-embedding-3	Flexible (e.g., 3072, user-defined)	Extremely high semantic accuracy, adaptable, well-documented API	Large-scale, accurate text retrieval and understanding
Cohere	1024, 4096	Optimized for business & enterprise NLP, highly multilingual	Chatbots, global business applications, and massive, diverse datasets
SentenceTransformer	Varies by model (e.g., 384, 768)	Open-source, highly customizable, efficient, great community support	Local deployment, specific tasks, developer experimentation
Titan Multimodal	Varies, supports multi-modal mapping	Integrates visual & textual embeddings, native multi-modal search	Cross-modal search (e.g., find images using text query), product catalog retrieval

Dimensions vs Cost Trade-off

High Dimensions

- High Semantic Accuracy
- Better Retrieval Quality



Low Dimensions

- Lower Latency
- Reduced Computing Cost

Example Dimensions:

1. 384 (Efficient & Fast)
2. 1024 (Balanced Performance)
3. 1536 (Maximum Precision)

Key Takeaways



RAG ENHANCEMENT

- Integrates external data to optimize LLM performance and ensure current context.



EMBEDDING VITALITY

- Critical for robust semantic understanding and accurate knowledge retrieval.



STRATEGIC TRADEOFFS

- Dimension selection (e.g., 384 vs 1536) must balance semantic accuracy against speed and computing cost.



MULTIMODAL UNIFICATION

- Unified embeddings enable cross-model capabilities like text-to-image and product catalog search.



Thank You!

Questions?

Happy to discuss RAG implementation & more